Classification of finitely generated modules over PID

Let R be a ring. A left R-module is an abelian group M equipped with an R-action on M:

$$R \times M \longrightarrow M$$
$$(a, m) \longmapsto a \cdot m$$

satisfying (1). $1_R \cdot m = m$

(2) $(r+s)\cdot m = r\cdot m + s\cdot m$

(3) $r\cdot(m+n) = r\cdot m + r\cdot n$

(4) $(r\cdot s)m = r\cdot(s\cdot m)$.

②. An R-submodule $N \subseteq M$ is an abelian subgroup of M.
s.t. $\forall r \in R,\ r\cdot N \subseteq N$.

Example. 1). $R = F$ a field $\Rightarrow$ F-vector spaces.

2). free modules $R^{\oplus n} = \{(a_1, \dots, a_n) \mid a_i \in R\}$.

$(a_1, a_2, \dots, a_n) + (b_1, \dots, b_n) = (a_1+b_1, \dots, a_n+b_n)$

$r(a_1, \dots, a_n) = (ra_1, \dots, ra_n)$

3). A $\mathbb{Z}$-module is an abelian group G. $\begin{cases} n\cdot g = g+\dots+g \\ (-n)\cdot g = -n\cdot g \end{cases}$

4). a left R-submodule of R $\Longleftrightarrow$ a left ideal of R

Def 3.2. ① Direct product: M_i $(i \in I)$ are left R-modules,
then $\prod_{i\in I} M_i = \{(m_i)_{i\in I} \mid m_i \in M_i\}$

② Direct sum: $\bigoplus_{i\in I} M_i = \{(m_i)_{i\in I} \mid m_i \in M_i$, all but finitely many m_i's are zero$\}$

Def 3.3. ① R-module homomorphism/isomorphism: (skip)

③. Definition of $\ker\phi$ & $\mathrm{Im}\,\phi$

③ quotient R-module: $M/N := \{m+N \mid m \in M\}$.

④. M is called irreducible iff the only submodule of M is 0 and M.

Remark. When R is a commutative ring.
for $\phi \in \mathrm{Hom}_R(M,N)$, $a \in R$, define $a\cdot\phi \in \mathrm{Hom}_R(M,N)$
by $(a\cdot\phi)(m) := a\cdot\phi(m) = \phi(am)$.
Hence $\mathrm{Hom}_R(M,N)$ is equipped with the structure of R-module
($(a\cdot\phi)(bm) = (b\cdot a\cdot\phi)(m) = (ba)\phi(m) = (ab)\phi(m)$)

Def 3.4 M a left R-module, $X \subseteq M$ a subset

Define $RX = \{a_1x_1 + \dots + a_nx_n \mid a_i \in R,\ x_i \in X\}$

(1) If $|X| < \infty$ then $N = RX$ is finitely generated by X

(2) If $\exists\, m \in M$ s.t. $N = Rm$, then $N \subseteq M$ is called cyclic.

(3) M is free of rank n iff $\exists\, x_1, \dots, x_n \in M$. $\forall m \in M$, $\exists!\ (a_1, \dots, a_n) \in R^{\oplus n}$ s.t. $m = a_1x_1 + \dots + a_nx_n$

Next we will prove the main theorems in this lecture.

Thm 3.5 Let R be a PID. Let N be a free R-module of rank n, and L a submodule of N.

Then (1) L is free of rank l ($l \le n$).

(2) There exists a basis $y_1, \dots, y_n$ of N so that $a_1y_1, \dots, a_ly_l$ is a basis of L and $a_1, \dots, a_l \in R \setminus \{0\}$ s.t. $a_1 \mid a_2$, $a_2 \mid a_3$, $\dots$, $a_{l-1} \mid a_l$

Lemma 3.6 Let R be an integral domain, and let N be a free R-module of n. Then any $(n+1)$ elements $x_1, \dots, x_{n+1} \in N$ are dependent linearly.

pf: (sketched) Consider the fraction field of R. □

proof of Thm 3.5. Consider the isomorphism $N \cong R^{\oplus n}$.

Consider the set $\{\phi(L) \mid \text{for all } \phi \in \mathrm{Hom}_R(N, R)\} = S$

It contains a maximal ideal (a_1) where $\phi_1 : N \to R$ with $\phi_1(y_0) = a_1$. ($\exists\, y_0 \in R$)

(Since R is a PID, hence Noetherian ring). (this proof is easy) · $I_1 \subsetneq I_2 \subsetneq \dots \subsetneq \dots$

consider $I = \bigcup_{n \ge 1} I_n$.

Then for any $\phi' \in \mathrm{Hom}_R(N, R)$ we have $a_1 \mid \phi'(y_0)$.

(In fact, $d = \gcd(a_1, \phi'(y_0)) \Rightarrow d = (r_1\phi + r_2'\phi')(y_0) \Rightarrow (d) \in S$ contradicting with maximality.) $(d) \supsetneq (a_1)$

Now we have $\pi_i(y_0) = a_1 b_i$ $(b_i \in R) \Rightarrow y_0 = (a_1b_1, \dots, a_1b_n)$

$\Rightarrow a_1\phi_0(b_1, \dots, b_n) = \phi(y_0) = a_1$ · let $y_1 := (b_1, \dots, b_n) \in N$.

$\Rightarrow \phi_0(y_1) = 1$.

Claim: (i) $N \cong Ry_1 \oplus \ker\phi$

(ii) $L = Ra_1y_1 \oplus (L \cap \ker\phi)$.

(i). $\forall x \in N$, we have $x = \underbrace{\phi(x) \cdot y_1}_{\in Ry_1} + \underbrace{x - \phi(x) \cdot y_1}_{\in \ker\phi}$.

$Ry_1 \cap \ker\phi = \{ry_1 \mid r\phi(y_1) = r = 0\} = \{0\}$

$\Rightarrow N \cong Ry_1 \oplus \ker\phi$.

(ii). $\forall x \in L$, we have $x = \underbrace{\phi(x) \cdot y_1}_{\phi(x) \in (a_1)} + \underbrace{(x - \phi(x) \cdot y_1)}_{\in L \cap \ker\phi}$

Back to the proof: $L = Ra_1y_1 \oplus (L \cap \ker\phi) = Ra_1y_1 \oplus Ra_2y_2 \oplus \dots \oplus Ra_{n-1}y_{n-1}$ (by Lemma 3.6),

thus we prove (1)

$y_1 \oplus Ra_1y_1 \oplus \dots \oplus Ra_{i-1}y_{i-1} \subseteq N = Ry_1 \oplus \dots \oplus Ry_n$ (by (1) & Induction)

suppose $a_1 \nmid a_2$ take $d = \gcd(a_1, a_2) = a_1r_1 + a_2r_2$ $(r_1, r_2 \in R)$

and homomorphism ϕ': $N \to R$

$y_1 \mapsto r_1$

$y_2 \mapsto r_2$

$y_i \mapsto 0$

then $\phi'(a_1y_1 + a_2y_2) = d \mid \phi$ then $a_1 \nmid \phi'(a_1y_1 + a_2y_2)$ contradiction with the construction in (1). So $a_1 \mid a_2$ similarly, we have $a_1 \mid a_2, \dots, a_{i-1} \mid a_i$ □.

Thm. 3.7. (Foundamental thm of finitely generated modules over PID's).

Let R be a PID and L a finitely generated R-module.

Then $L \cong R^{\oplus r} \oplus R/(a_1) \oplus \dots \oplus R/(a_m)$. $(a_i \mid a_{i+1})$, moreover, it is unique

pf: As L is finitely generated: $\exists x_i \in L$. ϕ: surjective, homomorphism

$\phi: R^{\oplus n} \twoheadrightarrow M$.

$(a_1, \dots, a_n) \longmapsto a_1x_1 + \dots + a_nx_n$

by the isomorphism theory of module: $M \cong R^{\oplus n}/\ker\phi$

$\exists$ basis $y_1, \dots, y_n \in R^{\oplus n}$. and $a_1, \dots, a_m \in R$. s.t. $R^{\oplus n} = Ry_1 \oplus \dots \oplus Ry_n$.

$\ker\phi = Ra_1y_1 \oplus \dots \oplus Ra_my_m$

$\Rightarrow M \cong \dfrac{Ry_1 \oplus \dots \oplus Ry_n}{Ra_1y_1 \oplus \dots \oplus Ra_my_m} \cong R^{\oplus(n-m)} \oplus R/(a_1) \oplus \dots \oplus R/(a_m)$

(the second isormorphism can be proved easily:

define φ: $Ry_1 \oplus \dots \oplus Ry_n \longrightarrow R^{\oplus(n-m)} \oplus R/(a_1) \oplus \dots \oplus R/(a_m)$.

$x \in Ry_i$ $(1 \le i \le m) \longmapsto \pi_i(x) + (a_i)$.

$x \in Ry_i$ $(m+1 \le i \le n) \longmapsto \pi_i(x)$

$\Rightarrow \ker\varphi = Ra_1y_1 \oplus \dots \oplus Ra_my_m$)

Next we will prove the hardest part: uniqueness.

by Chinese remainder theorem:

$M = R^{\oplus r} \oplus R/(p_1^{\alpha_{1,1}}) \oplus \dots \oplus R/(p_1^{\alpha_{1,s_1}}) \oplus R/(p_2^{\alpha_{2,1}}) \oplus \dots$

where $a_1 = p_1^{\min\{\alpha_{1,1}, \dots, \alpha_{1,s_1}\}} p_2^{\min\{\alpha_{2,1}, \dots, \alpha_{2,s_2}\}} \dots$

...

We want to show that: r, p_i, $\alpha_{i,j}$ can be determined by M itself!

Lemma 3.8. p, q are prime elements such that $(p) \neq (q)$.

then (1) If $M = R$. then $p^mM/p^{m+1}M \cong R/(p)$

(2) If $M = R/(p^n)$, then $p^mM/p^{m+1}M \cong \begin{cases} R/(p) & \text{if } m < n \\ 0 & \text{if } m > n \end{cases}$

(3) If $M = R/(q^n)$. then $p^mM/p^{m+1}M = 0$.

Back to the proof we say that:

$$\dim_{R/(p_i)} \left(M / p_i M \right) = r + \#\{\alpha_{i,1}, \dots, \alpha_{i,s_i}\}$$

$$\dim_{R/(p_i)} \left(pM / p_i^2 M \right) = r + \#\{\alpha_{i,j} \geq 2\}.$$

Let p varying all prime elements in R and all $m \in \mathbb{Z}_{\geq 0}$ we can determine all the numbers $r, p_1, \dots p_r, \alpha_{i,1}, \dots$ $\square$